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COURSE: NETWORK APPLICATION DEVELOPMENT

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STARTED DATE PROJECT: 16th of June

FINISHED DATE PROJECT: 18th of June

STUDENT ID: 45559

MY NETWORK APPLICATION PROJECT: **LinguaGlobal Academy (Language Academy Dashboard)**

THE PROJECT IS A LANGUAGE ACADEMY (LINGUA) - WHICH INVOLVES LEARNING MULTIPLE LANGUAGES, THE COST OF LEARNING THEM, TO TUTOR THE PEOPLE/STUDENTS/USERS WITH THE LANGUAGE LEVEL OF WHAT IT REQUIRES AND TO LEARN MORE WORDS, SENTENCES, TO KNOW HOW TO SPEAK TO OTHER PEOPLE IN ANY LANGUAGE THEY WANT TO LEARN AND GAIN KNOWLEDGE, AND THE DASHBOARD OF THE AMOUNT OF THE PRICES, PEOPLE, PERCENTAGES, ETC.

MY WORK:

<http://localhost:62906/Default.aspx>

STEPS ON HOW I DID MY MINI LINGUAGLOBAL ACADEMY LANGUAGE ACADEMY DASBOARD:

1. I opened Microsoft Visual Studio
2. Created new project in Microsoft VS
3. I had chosen Web Service Application (+C)

4. Then I named my file of the project:(language academy service) and chose with **(.NET Framework 4.8.)**
5. Then it had opened the file for to create the project
6. At my side, there was a solution language academy service and below that there was (Iservice1.cs) and double clicked on it
7. I had after erased the old code which was not usable and had created a new one to make the application work
8. (Iservice1.cs) - defines inside the user profile, dashboard numbers, the prices, and language grades
9. The other step – is to click the little arrow next to (Service1.svc) and it will reveal (Service.svc.cs), and I had doubled clicked on it and did the same exact task as (Iservice1.cs) with a new code and started to build it to make it soon to run the application dashboard
10. (Service.svc.cs) - defines inside the all the names, descriptions, prices, and the dashboard numbers
11. Before making the client interface, I had decided to make that my backend will work, and it will run flawlessly
12. Started the green button on top of the visual studio, then the WCF Test Client will pop up, double clicked the (GetAcademymyDetails()),(GetAcademymyprofile()), (GetAcademymyDashboard()), clicked the invoke button, then the pricing, names, description, numbers will appear in response to the section down below without any errors and then close that window to stop debugging
13. We go to the top menu bar of VISUAL STUDIO and click: project – add new item and in that list, you select from them the WEB FORM and you click on it and name it exactly: (Default.aspx) and click add
14. Then you double click on it and do the same exact previous 2 tasks which was done
15. (Default.aspx) - it builds inside the home page layout, profile card, metric dashboard cards, and a clean CSS-based bar graph
16. Click on the small expansion arrow right next to (Default.aspx) to see it's hidden file which is called (Default.aspx.cs)
17. Double click on it and repeat doing the same routine just as i had on these last tasks
18. (Default.aspx.cs) - it quires inside the WCF data which will instantly upon page upload and draws the dashboard bars
19. Right click on the (Default.aspx), and choose to view it in browser on the right
20. Select as start page
21. And then you press the green start button at the top of your visual studio interface
22. And it will launch and be shown the web system layout all with everything it is needed like: required language information, pricing blocks, student data, and

colorful live tracking visual graph which is mapped straight from your backend WCF architecture and the end.

ALL CODES:

1. (Iservice1.cs):

```
using System;
```

```
using System.Collections.Generic;
```

```
using System.Runtime.Serialization;
```

```
using System.ServiceModel;
```

```
namespace LanguageAcademyService5
```

```
{
```

```
    [ServiceContract]
```

```
    public interface IService1
```

```
    {
```

```
        [OperationContract]
```

```
        AcademyInfo GetAcademyDetails();
```

```
        [OperationContract]
```

```
        StudentProfile GetStudentProfile();
```

```
        [OperationContract]
```

```
DashboardData GetDashboardData();  
}
```

```
[DataContract]  
public class AcademyInfo  
{  
    [DataMember] public string Name { get; set; }  
    [DataMember] public string Description { get; set; }  
    [DataMember] public string Importance { get; set; }  
    [DataMember] public string HowItWorks { get; set; }  
    [DataMember] public string MonthlyPrice { get; set; }  
    [DataMember] public string YearlyPrice { get; set; }  
}
```

```
[DataContract]  
public class StudentProfile  
{  
    [DataMember] public string FullName { get; set; }  
    [DataMember] public string StudentID { get; set; }  
    [DataMember] public string EnrolledProgram { get; set; }  
    [DataMember] public string Level { get; set; }  
}
```

```
[DataContract]  
public class DashboardData  
{
```

```

[DataMember] public int TotalHoursLearned { get; set; }

[DataMember] public int CompletedLessons { get; set; }

[DataMember] public double OverallProgressPercentage { get; set; }

[DataMember] public Dictionary<string, int> LanguageGrades { get; set; }

}

}

```

2. (Service.svc.cs):

```

using System;

using System.Collections.Generic;

namespace LanguageAcademyService5
{
    public class Service1 : IService1
    {
        public AcademyInfo GetAcademyDetails()
        {
            return new AcademyInfo
            {
                Name = "LinguaGlobal Academy",
                Description = "A premier language learning platform designed for immersive tutoring.",
                Importance = "Fluency opens global career opportunities and bridges cultural gaps.",
                HowItWorks = "We pair you with certified native speakers for interactive 1-on-1 online sessions.",
                MonthlyPrice = "$49 / month",
            }
        }
    }
}

```

```
        YearlyPrice = "$450 / year (Save 25%)"
    };
}

public StudentProfile GetStudentProfile()
{
    return new StudentProfile
    {
        FullName = "John Doe",
        StudentID = "LA-2026-8941",
        EnrolledProgram = "Dual-Language Immersive Track",
        Level = "Intermediate (B2)"
    };
}
```

```
public DashboardData GetDashboardData()
{
    return new DashboardData
    {
        TotalHoursLearned = 124,
        CompletedLessons = 48,
        OverallProgressPercentage = 78.5,
        LanguageGrades = new Dictionary<string, int>
        {
            { "Spanish (B2)", 88 },
            { "French (A1)", 92 },
        }
    }
}
```

```

        { "German (A2)", 75 }
    }
};
}
}
}

```

3. (Default.aspx):

```

<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
Inherits="LanguageAcademyService5.Default" %>

```

```

<!DOCTYPE html>

```

```

<html xmlns="http://www.w3.org/1999/xhtml">

```

```

<head runat="server">

```

```

    <title>Language Academy Dashboard</title>

```

```

    <style>

```

```

        body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background-
color: #f4f6f9; margin: 0; padding: 20px; color: #333; }

```

```

        .container { max-width: 1100px; margin: 0 auto; }

```

```

        .header { background: #4A154B; color: white; padding: 20px; border-radius: 8px;
margin-bottom: 20px; text-align: center; }

```

```

        .grid { display: grid; grid-template-columns: 2fr 1fr; gap: 20px; }

```

```

        .card { background: white; padding: 25px; border-radius: 8px; box-shadow: 0 4px 6px
rgba(0,0,0,0.05); margin-bottom: 20px; }

```

```

        h2 { color: #4A154B; border-bottom: 2px solid #f4f6f9; padding-bottom: 10px; margin-
top: 0; }

```

```
.price-tag { display: inline-block; background: #E01E5A; color: white; padding: 8px 15px; border-radius: 20px; font-weight: bold; margin-right: 10px; }
```

```
.metric-box { display: inline-block; background: #f4f6f9; padding: 15px; border-radius: 6px; text-align: center; min-width: 100px; margin-right: 15px; }
```

```
.metric-num { font-size: 24px; font-weight: bold; color: #4A154B; }
```

```
.bar-container { background-color: #e0e0e0; border-radius: 12px; margin: 15px 0; position: relative; height: 25px; }
```

```
.bar-fill { background-color: #2EB67D; height: 100%; border-radius: 12px; text-align: right; line-height: 25px; color: white; padding-right: 10px; font-weight: bold; font-size: 13px; }
```

```
.subject-row { margin-bottom: 15px; }
```

```
.subject-name { font-weight: bold; margin-bottom: 5px; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<form id="form1" runat="server">
```

```
<div class="container">
```

```
<div class="header">
```

```
<h1><asp:Label ID="lblAcademyName" runat="server"></asp:Label></h1>
```

```
<p>Your Gateway to Global Communication</p>
```

```
</div>
```

```
<div class="grid">
```

```
<div>
```

```
<div class="card">
```

```
<h2>About Our Academy</h2>
```

```
        <p><strong>What it is:</strong> <asp:Label ID="lblDesc"
runat="server"></asp:Label></p>
```

```
        <p><strong>Why it matters:</strong> <asp:Label ID="lblImportance"
runat="server"></asp:Label></p>
```

```
        <p><strong>How it works:</strong> <asp:Label ID="lblHow"
runat="server"></asp:Label></p>
```

```
    </div>
```

```
<div class="card">
```

```
    <h2>Student Learning Dashboard</h2>
```

```
    <div style="margin-bottom: 25px;">
```

```
        <div class="metric-box">
```

```
            <div class="metric-num"><asp:Label ID="lblHours"
runat="server"></asp:Label></div>
```

```
            <div>Hours Studied</div>
```

```
        </div>
```

```
        <div class="metric-box">
```

```
            <div class="metric-num"><asp:Label ID="lblLessons"
runat="server"></asp:Label></div>
```

```
            <div>Lessons Finished</div>
```

```
        </div>
```

```
        <div class="metric-box">
```

```
            <div class="metric-num"><asp:Label ID="lblProgress"
runat="server"></asp:Label>%</div>
```

```
            <div>Total Progress</div>
```

```
        </div>
```

```
</div>
```

<h3>Subject Course Results (Grades)</h3>

<asp:PlaceHolder ID="phGradesGraph" runat="server"></asp:PlaceHolder>

</div>

</div>

<div>

<div class="card">

<h2>Student Profile</h2>

<p>Name: <asp:Label ID="lblStudentName" runat="server"></asp:Label></p>

<p>ID: <asp:Label ID="lblStudentID" runat="server"></asp:Label></p>

<p>Program: <asp:Label ID="lblProgram" runat="server"></asp:Label></p>

<p>Current Status: <asp:Label ID="lblLevel" runat="server"></asp:Label></p>

</div>

<div class="card">

<h2>Academy Tuition Plans</h2>

<p>Standard Month-to-Month Access:</p>

<asp:Label ID="lblMonthPrice" runat="server"></asp:Label>

<p style="margin-top: 20px;">Annual Discounted Membership:</p>

<asp:Label ID="lblYearPrice" runat="server"></asp:Label>

</div>

</div>

```
</div>

</div>

</form>

</body>

</html>
```

4. (Default.aspx.cs):

```
using System;

using System.Web.UI;

using System.Web.UI.WebControls;

namespace LanguageAcademyService5
{
    public partial class Default : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack)
            {
                try
                {
                    // Directly instantiate our WCF service class internally to read configuration data
                    smoothly
                    Service1 service = new Service1();
```

```
// 1. Fetch and bind Academy Presentation Text
var academy = service.GetAcademyDetails();
lblAcademyName.Text = academy.Name;
lblDesc.Text = academy.Description;
lblImportance.Text = academy.Importance;
lblHow.Text = academy.HowItWorks;
lblMonthPrice.Text = academy.MonthlyPrice;
lblYearPrice.Text = academy.YearlyPrice;

// 2. Fetch and bind Student Profile Information
var profile = service.GetStudentProfile();
lblStudentName.Text = profile.FullName;
lblStudentID.Text = profile.StudentID;
lblProgram.Text = profile.EnrolledProgram;
lblLevel.Text = profile.Level;

// 3. Fetch Metrics and render Dynamic Dashboard HTML Graphs
var dashboard = service.GetDashboardData();
lblHours.Text = dashboard.TotalHoursLearned.ToString();
lblLessons.Text = dashboard.CompletedLessons.ToString();
lblProgress.Text = dashboard.OverallProgressPercentage.ToString("0.0");

// Build the language bar graphs natively from the dictionary
foreach (var subject in dashboard.LanguageGrades)
{
```

```

        string subjectHtml = $"
            <div class='subject-row'>
                <div class='subject-name'>{subject.Key} — Current Grade:
{subject.Value}%</div>
                <div class='bar-container'>
                    <div class='bar-fill' style='width:
{subject.Value}%;'>{subject.Value}%</div>
                </div>
            </div>";

        phGradesGraph.Controls.Add(new LiteralControl(subjectHtml));
    }
}
catch (Exception ex)
{
    lblAcademyName.Text = "System Error loading dashboard data: " + ex.Message;
}
}
}
}
}
}
}

```

